



RetailPulse – AI-Powered Customer Analytics & Demand Forecasting Platform

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Data Analytics Internship Project

1. Introduction

Retail Pulse AI is a data analytics project designed to analyze retail sales data and generate actionable insights for business decision-making.

2. Problem Statement

Businesses struggle to understand sales patterns, customer behavior, and future demand trends.

3. Objectives

- Analyze sales and profit
 - Identify top-performing categories and regions
 - Segment customers
 - Predict future sales
 - Build interactive dashboard
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4. Dataset

- Sales data
 - Columns: sales, profit, region, category, city, discount, quantity
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5. Methodology

- Data preprocessing
- Feature engineering
- Exploratory Data Analysis

- Machine Learning (KMeans, XGBoost)
 - Time series forecasting (Prophet)
 - Dashboard development (Streamlit)
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6. Key Results

- Identified top regions and categories
 - Customer segmentation achieved using clustering
 - Forecast model predicts next 30 days sales
 - Dashboard enables real-time filtering
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7. Conclusion

Retail Pulse AI successfully transforms raw retail data into meaningful insights and predictive analytics.

8. Future Scope

- Add real-time database
- Improve ML models
- Deploy on cloud (AWS/Streamlit Cloud)